

Nikhil Prabhu

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EDUCATION

University of California, San Diego (UCSD)

B.S. in Computer Science — Cumulative GPA: 3.67

San Diego, CA

Expected Graduation: June 2028

SKILLS

Languages: Python, Java, JavaScript, TypeScript, Dart **Frameworks:** Next.js, Node.js, Flask, FastAPI, Electron
Tools: Git, Docker, Bash, Vercel, Fly.io **Concepts:** Machine Learning, Computer Vision, OCR, LLM Fine-tuning

EXPERIENCE

UCSD Agent Development (Clue2)

San Diego, CA

Desktop Agent Developer — Founder

Jan 2025 – Present

- Architected and developed a cross-platform desktop agent using Electron and TypeScript that processes live meetings and autonomously executes tasks across user applications.
- Engineered a real-time audio transcription pipeline using AssemblyAI WebSocket streaming with dual-channel audio capture, buffering, and fault-tolerant session management.
- Built an LLM-driven workflow automation engine translating natural language commands into structured API calls across 7+ services including Gmail, Google Calendar, Docs, Sheets, Slack, and Notion.
- Working under mentorship of Professor Ranmoham Pathuri with a team of 5 PhD researchers; scaled a beta version to 40+ users, with early adoption by Y Combinator-backed startups.

Undergraduate Research Assistant — AI for Circuit Invention

San Diego, CA

Research Assistant

March 2026 – Present

- Developing an LLM-based pipeline for automated generation and verification of novel analog circuit topologies
- Integrating LLM APIs with Cadence simulation tools to evaluate AI-generated designs against state-of-the-art benchmarks
- Collaborating with Samvid AI and Prasad Gudem on reinforcement learning strategies for circuit design space exploration

Foot Drop Ankle-Foot Orthosis Research

Atlanta, GA

Lead Researcher – IEEE ICHCI Presenter

May 2024 – Nov 2024

- Conducted research at Georgia Tech's mmWave Antennas Laboratory, developing a 3D-printed ankle-foot orthosis controlled via electromyography signals.
- Coded custom motor actuator system, improving gait response precision by 23%.
- Refined the orthosis prototype through and biomechanical modeling in collaboration with Georgia Tech PhD researchers.
- Presented and published findings at the 2024 IEEE ICHCI Conference (Impact Factor 3.45) as youngest accepted researcher.

Brain-Computer Interface (BCI) Device Research

Stanford, CA

Lead Researcher – Stanford Lee-Messer Lab, School of Medicine

Oct 2023 – Dec 2024

- Initiated and led development of a brain-computer interface at Stanford's Lee-Messer Lab to help motor-impaired students communicate using neural signals.
- Built and benchmarked ML algorithms using P300 paradigm to convert EEG signals into tablet commands
- Developed React Native frontend integrating BCI control for assistive learning platform, piloted testing with 12 patient users.

PROJECTS

Tuberculosis Genomic Co-Resistance Research

Remote

Lead Researcher – IEEE ICBT 2025 Presenter

Dec 2024 – May 2025

- Analyzed genomic data from 2,200+ *Mycobacterium tuberculosis* samples using k-mer analysis and machine learning to model co-resistance between Rifampicin and Isoniazid.
- Developed predictive model achieving 86% accuracy; accepted for publication and presentation at IEEE International Conference on Bioinformatics and Biomedicine Technologies.

CaféCode – TypeScript Web Application

Remote

Full-Stack Developer

May 2025 – Sept 2025

- Architected and maintained a full-stack TypeScript web app deployed on Vercel; implemented modular React/Next.js frontend, and serverless API Routes.
- Optimized production performance via code-splitting, lazy loading, and type-safe data flows.